LESSON 3.5
USING OPERATIONS TO SOLVE REAL-WORLD PROBLEMS – PART 2

LESSON INTRODUCTION

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

LEARNING TARGETS

- I can solve one-step, real-world problems involving any of the four operations with decimals or fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.

WORKING TOWARD

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What financial terms are associated with the four operations?
- How do you know which operation to use in real-world problems?

LESSON NARRATIVE

In this lesson, students solve one-step problems involving money with decimals in a real-world finance context. In the guided sections, students learn about financial terms and complete activities that form deeper problem-solving connections and strengthen operational fluency using real-world situations. Students then use this practice to complete one-step problems involving real-world finance situations.

COMPONENT SUMMARY

3.5.1 WARM-UP *(~10 MIN)*

Budgeting video and discussion

3.5.2 GUIDED INSTRUCTION *(20-25 MIN)*

Guided introduction to financial terms and real-world problems

3.5.3 GUIDED PRACTICE *(15-20 MIN)*

Practice completing one-step problems involving real-world finance situations

3.5.4 WRAP-UP *(~5-10 MIN)*

Move on to solving one-step problems involving real-world finance situations

RESOURCES

GLOSSARY

Key Terms:

- Income
- Finance
- Investment
- Principal
- Return
- Gross income
- Net income

Additional Terms:

- N/A

MATERIALS

- Budgeting Video
- (Optional) Chart paper
- (Optional) Markers
- (Optional) Frayer Models for vocabulary

ADDITIONAL PREPARATION

- 3.5.1 Warm-Up contains a video to accompany the question prompts.
- (Optional): Create table for problems in 3.5.3 Guided Practice.
- (Optional): Print or prepare students to create Frayer Models or an organized table for the vocabulary words in this lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse financial terms with their mathematical impact or meaning, like investment and return.

THINGS TO CONSIDER

- Consider "pre-work" for this lesson as the previous day's homework to better prepare students to navigate financial terms for this lesson (students can watch a video explaining these terms as "pre-work" or completing question 3 in 3.5.2 as "pre-work" for the lesson using an online dictionary).
- Consider providing laminated sample budgets, checks, and paychecks for students to reference during the lesson. Circle values that align with the terms.

LITERACY AND LANGUAGE ROUTINES

Three Reads

3.5.2 Guided Instruction provides numerous opportunities for students to utilize reading comprehension skills in a real-world mathematic scenario. Consider applying "Three Reads" routines throughout the activity, with an emphasis on problems with higher complexity. Students can also write new financial terms, definitions, examples, and non-examples in journals following the "Three Reads".

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

Students will be introduced to financial terms that may be unfamiliar. Encourage students to look for similarities and patterns when solving real-world finance problems that focus on relevant words/phrases, creating a plan to solve, and relate financial terms to concepts in mathematical operations. Students can connect strategies learned in Lesson 4 to the finance-specific problems of this lesson.

DIFFERENTIATE INSTRUCTION

Students are introduced to potentially new financial terms. Consider having students create a graphic organizer, like a table of Frayer Models for each vocabulary words, from 3.5.3 Guided Practice. The table can be drawn in math journals or on chart paper with the word, operation, examples, and non-examples, as a visual aid for problem scaffolding.

Consider a variety of methods for students to demonstrate or express their learning, using projects like budget sheets, balancing the payroll, small business CEO, and more to engage students in multiple representations of their learning.

3.5.1 WARM-UP: TWO QUESTIONS AND ONE WONDER

***Component Overview:** Display the Warm-Up 4-minute video for the entire class. In the video, Ky-ree, a firefighter in Los Angeles, discusses budgeting his salary and monthly paycheck.*

TEACHER MOVES

Utilize this discussion as an opportunity to begin incorporating the math vocabulary of this lesson in context, even referencing or beginning to fill in the Word Bank in 3.5.2 Guided Instruction to orient students to the content.

QUESTIONING

As an extension for project-based learning, consider asking other adults with various types of jobs (or have students research other salaries) to share an example of their monthly budget. These budgets can be laminated and used by students to help them get familiar with the vocabulary.

3.5.1 Warm-Up: Two Questions and One Wonder

Ky-ree, a firefighter in Los Angeles, budgets his paycheck.

Ky-ree's Budget	
Monthly salary	$5,800
Taxes and other deductions	$2,900
Net take home income	$2,900
Rent	$1,350
Utilities	$60
Internet & cable	$50
Cell phone	$130
Car payment	$330
Insurance	$120
Gas	$140
Food	$250
Gym	$30
Entertainment	$120
Remaining	$320

1. **List two questions you have about Ky-ree's budget.**
   Answers will vary. What do utilities include? What about water and electric?

2. **List one thing you wonder about Ky-ree's budget.**
   Answers will vary. I wonder what is being added or subtracted.

3.5.2 Guided Instruction: Real-World Contexts About Finances

Income is the amount of money received.

1. Recall the problem from Lesson 4.
A Valencia orange tree that is 12 years old can produce 1.4 boxes of fruit. In May 2017, a box of fresh Valencia oranges cost \$20.45. How much income would a Valencia orange tree generate?
\$28.63 of income

Finance is the management of money.

2. Daniel owns a Valencia orange grove. Choose other items from the list that he should consider as a part of his finances.
   - The taxes he pays on the property.
   - The amount of money he has in the bank.
   - The cost of taking care of the orange grove.

3. Use the words in the word bank to complete each statement.

Word Bank

withdrawal	checking	costs	deposit
expenses	savings	taxes	

a. *Costs* or *expenses* describe money that is spent.

b. *Taxes* are the amount of money that is paid to the government.

c. A *deposit* is the amount of money placed in a bank account.

d. A *withdrawal* is the amount of money removed from a bank account.

e. *Checking* and *savings* are types of bank accounts.

An ***investment*** is when the initial amount of money, known as the ***principal***, earns more money, known as the ***return***, over time.

4. For each financial transaction, write the mathematical operation that is most commonly associated with it.

Financial Transaction	Mathematical Operation
costs	−
deposit	+
expenses	−
income	+
taxes	−
withdrawal	−
return on an investment	+

Gross income is the amount of income that is earned.

5. Hannah earns \$9.17 an hour. Each week she works 14 hours. How much is Hannah's weekly gross income?
\$128.38 per week

Net income is the amount of income that is earned after expenses and taxes are deducted.

6. Each week Hannah has to pay \$26.96 in taxes. What is Hannah's net income?
\$101.42 per week

3.5.2 GUIDED INSTRUCTION: REAL-WORLD CONTEXTS ABOUT FINANCES

Component Overview: *Students are introduced to financial terms and complete activities designed to provide opportunities to form deeper connections using real-world situations.*

TEACHER MOVES

Three Reads

Consider modeling the Three Reads routine for students to later apply in Guided Practice, targeting the interpretation and translation of verbal to algebraic descriptions.

- First Read: Reading through the prompt to get familiar with the context.
- Second Read: Reading through the prompt specifically looking for important information to circle, underline, or note.
- Third Read: Read the question and look for information within the prompt to answer the question.

DIFFERENTIATE INSTRUCTION

As financial terms, definitions, and examples are introduced, consider having students read and write the definitions in a math journal using a table and/or Frayer Model.

DIFFERENTIATE INSTRUCTION

Consider placing students in groups of 2 to 3 to complete question 3 and discuss options. Encourage small groups to discuss how and why they chose the words they did to fill in the blanks. Ensure that the students have chosen the correct terms, as a firm understanding of these finance terms is important to problem solving in upcoming activities.

DIFFERENTIATE INSTRUCTION

Consider reproducing this table and adding other key terms as an anchor chart for interpreting mathematical phrases in real-world contexts for students to use throughout this unit. Encourage a variety of depictions, including verbal descriptions, drawn pictures, printed images, and more.

3.5.3 GUIDED PRACTICE: SOLVING REAL-WORLD FINANCE PROBLEMS

Component Overview: *Students solve one-step problems involving real-world finance situations. Consider working as a class for question 1 to model a "Think Aloud" and how to use visual diagrams to support problem solving, then having students work independently or in small groups for questions 2–4. Remind students to use the table and definitions in previous sections to work out the problems.*

DIFFERENTIATE INSTRUCTION

Consider providing a pre-constructed table for scaffolding practice or encouraging students to model their own strategies for making sense of the concepts when practicing MA.K12.MTR.7.1.

Example:

WHAT IS THE QUESTION ASKING?	OPERATION (±×÷)	MY APPROACH / STRATEGY

DIFFERENTIATE INSTRUCTION

For a visual model, in question 1d, draw a bar diagram that shows the total costs in sections, including the unknown of the net income that is determined through subtraction.

TEACHER MOVES

Three Reads

As students work in groups to work on questions 2-4, have students use Three Reads to make sure they understand what they are trying to find in relation to financial terms. Students can highlight, circle, or underline vocabulary words that they notice.

QUESTIONING

Challenge students to create new financial problems that use the information given in the existing problems or from Ky-ree's budget in the Warm-Up. The new problems can be one step and real-world, then have students work in small groups to solve the problems.

3.5.3 Guided Practice: Solving Real-World Finance Problems

1. A trampoline park charges $26.88 for 2 hours.
   a. How much is the cost for 1 hour?
   $13.44 *per hour*
   b. The manager of the trampoline park knows that during a Saturday there are typically 125 people at the park. If each person only pays for 2 hours, what is the park's typical income on Saturdays?
   $3,360.00
   c. The manager of the trampoline park lists the costs for Saturday as follows: $119.50 to pay the employees and $256.73 for all other costs. What are the park's typical costs?
   $376.23
   d. Determine the park's net income for a typical Saturday.
   $2,983.77
2. Emmanuel invested $12,500 into the stock market. In the first quarter of the year his return was $281.25. He combines the return with his initial investment. How much does Emmanuel have invested after the first quarter of the year?
   $12,781.25
3. Aaron has $2,478.23 in his checking account. He writes a check for $637.89 to pay for new tires on his truck. How much money does he have in his checking account now?
   $1,840.34
4. Violeta has $5,734.49 in her savings account. She deposits $218.56. How much does she have in her savings account now?
   $5,953.05

3.5.4 Wrap-Up: Ticket Out the Door

Mishka is planning to host a music festival to raise funds for a local food bank. Her budget is shown.

Item	Amount
Ticket price at the door	$9.00
Fee paid to each band	$175.00
Early-bird ticket price	$6.50
Equipment rental	$225.00
Publicity (flyers/artwork)	$64.57
Stage rental	$187.50

1. In order to find the **gross income** for the music festival, I need to know *the number of people who bought an early bird ticket* and *the number of people who bought a ticket at the gate*.

2. List the items in Mishka's budget that are **expenses**.
Band fee, equipment rental, publicity, and stage rental

3. If Mishka had a gross income of $1,166.50 and expenses totaling $827.07, what is her net income for the music festival?
$339.43

3.5.4 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students work individually to use a given budget for a music festival to answer a series of questions and solve a problem related to the planning. Student understanding of financial terms learned in this lesson is also assessed through embedded questioning. Students should not use a calculator to complete the Wrap-Up.*

QUESTIONING

Challenge students by incorporating new financial terms like revenue and profit to re-word problems from the activities in the lesson.

Example: After defining revenue and profit, ask, "How are gross income and net income related to revenue and profit?" or "Write your own word problem about the Valencia Grove using key terms from the lesson."